Here's a **comprehensive expanded list (50+) of online & offline tools** that can help you identify **AI-generated content** — including **text, code, images, multimedia, and plagiarism-related detection**. The table includes tools, what they detect, core features, and quick notes on usage or pricing where available.

 Important: No detector is perfectly accurate — combine tools and human review for high-stakes work. (Medium)

 Expanded List: Tools for Detecting AI-Generated or Non-Original Content

Tool/Software	Detects	Type	Key Features	Notes
Audio Forensics Tools	Audio	Offline	Voice synthesis detection	Research & lab focus
CodeMatch	Code	Online/ Offline	Code reuse detection	Static analysis
Blackbox AI Detector	Code	Online	AI code detection	Basic signals
DroidDetect (research)	Code	Offline/ Research	AI code detection model	Experimental suite (arXiv)
MOSS	Code similarity	Online	Code similarity analysis	Classic academic use
JPlag	Code similarity	Online/ Download	Detects code similarity	Supports many languages
Codequiry	Code similarity	Online	AI + copied code detection	Commercial code checks
Deepware Monitor	Deepfake	Online	Continuous monitoring	Enterprise
Deepware Scanner	Deepfake Video	Online	Video deepfake detection	Specialized for video
Sensity AI	Deepfakes	Online	Detects Deepfake/AI video	Enterprise video analysis

Winston OCR AI Detector	Documents	Online	Scan PDFs / handwriting	OCR + AI detection (GPTinf)
Meta AI-labeling	Images	Platform feature	Official AI-content labels	Evolving rollout (Lifewire)
AI Image Detector (DDIY list)	Images	Online	AI-image identification tools	Useful for images (Don't Do It Yourself)
FotoForensics	Images	Online/ Offline	Pixel-level forensic analysis	Good for tampered images
Exposing.ai	Images	Online	Photo provenance and AI trace search	Dataset cross-checking (guides.library.ttu.edu)
Google SynthID (via Gemini)	Images, Video, Audio	App/ Online	Detects AI content via watermarks	New multimodal detection (The Verge)
AI Forensic Labs	Multimedia	Offline/ Enterprise	Deep analysis	Research labs
Turnitin AI	Text	Online	AI-writing detection + plagiarism, LMS integration	Institutional / paid (ASU Prep Global)
GPTZero	Text	Online	Sentence-level AI score, free/paid	High accuracy for AI text (GPTZero)
Originality.ai	Text	Online	AI + plagiarism detector	Cloud API & reports (Originality.ai)
QuillBot AI Detector	Text	Online	Detects AI/AI-refined sentences	Free & paid options (QuillBot)
ZeroGPT	Text	Online	Free AI text scan	Basic AI detection (SEO.com)
Scribbr AI Detector	Text	Online	Free & premium AI text detection	Used by academic writers (Scribbr)
Sapling AI Detector	Text	Online	Free / SEO use	Simple AI/grammar check (Zapier)
Smodin	Text	Online	AI detector + paraphrase	Student-friendly tool (Jotform)

Monica AI Detector	Text	Online	Multi-source AI review	Aggregates signals (Jotform)
StoryChief AI Detector	Text	Online	Free detector	Included in content tools (StoryChief)
Surfer AI Content Detector	Text	Online	SEO-oriented detector	Lightweight analysis (StoryChief)
NoteGPT AI Detector	Text	Online	Free AI check	Quick scanning tool (NoteGPT)
Detecting-AI	Text	Online	User-friendly AI indicator	Simple indicator tool (Medium)
GPT-2 Output Detector	Text	Online	Classic AI text classifier	Older model still in use (Scribbr)
OpenAI Text Classifier (historic)	Text	Online	AI detection from OpenAI	Tool availability varies (Scribbr)
Plagiarism Checker (Grammarly)	Text	Online	Plagiarism + some AI alerts	Popular writing tool (Grammarly)
Grammarly Editor	Text	Online	Grammar + AI detection	Helps refine & flag AI
Copyscape	Text	Online	Plagiarism detection	Can flag copied/generated text
CrossPlag	Text	Online	AI + plagiarism	Academic use (Scribbr)
Turnitin iThenticate	Text	Online	Publisher plagiarism service	Professional publishing
PlagScan	Text	Online	Plagiarism + AI markers	Academic reports
Turnitin WriteCheck	Text	Online	Writing analysis	Educator tool
Turnitin Revision Assistant	Text	Online	Instructor feedback	Helps identify AI scribes

Pangram Labs AI Detector	Text	Online	Tested among top detectors	High accuracy in tests (pangram.com)
Sapling Grammar + AI	Text	Online	Grammar + AI flags	Business teams
Grammarly Plagiarism + AI	Text	Online	Combined detection	Widely used
Undetectable AI	Text	Online	Humanizes AI text	Not detector but bypass tool (Medium)
Phrasly.AI (humanizer)	Text rewriter	Online	Edits to evade detectors	Complementary tool (Medium)
Copyleaks	Text, Images	Online/ API	AI detector, plagiarism, multi-lang	Expanding to image detection (Copyleaks)
AI Detector Tools Directory (Detectortools)	Text, Multimedia	Online	Search portal of detectors	Central lookup list (guides.library.ttu.edu)
Winston AI	Text, Plagiarism	Online	AI detector + plagiarism + visual reports	Education & SEO focus (Winston AI)
Microsoft Video Authenticator	Video/ Images	Offline/ Online	Flags AI/morphed content	Research prototype
Academic Keystroke AI Detector	Writing behavior	Offline/ Research	Behavioral AI signal detection (arXiv)	

□ Categories & When to Use Them

Text / Document AI Detection

Use these for essays, articles, SEO content, academic submissions:

- **GPTZero, QuillBot AI Detector, Scribbr, ZeroGPT, Copyleaks, Winston AI** — text probability and AI patterns. (SEO.com)

Code Detection

Use for code similarity or AI-assisted code:

- **MOSS, JPlag, Codequiry, DroidDetect (research)** — structural code comparison. (arXiv)

Image / Multimedia Detection

Use for photos, artwork, deepfakes:

- **Google SynthID, Copyleaks image detection, Exposing.ai, Deepware, Sensity.** (The Verge)

Plagiarism + Combined Detection

If you want both copying and AI signals:

- **Turnitin suite, Copyleaks, PlagScan, Grammarly, CrossPlag.** (Grammarly)

Research / Cutting-edge

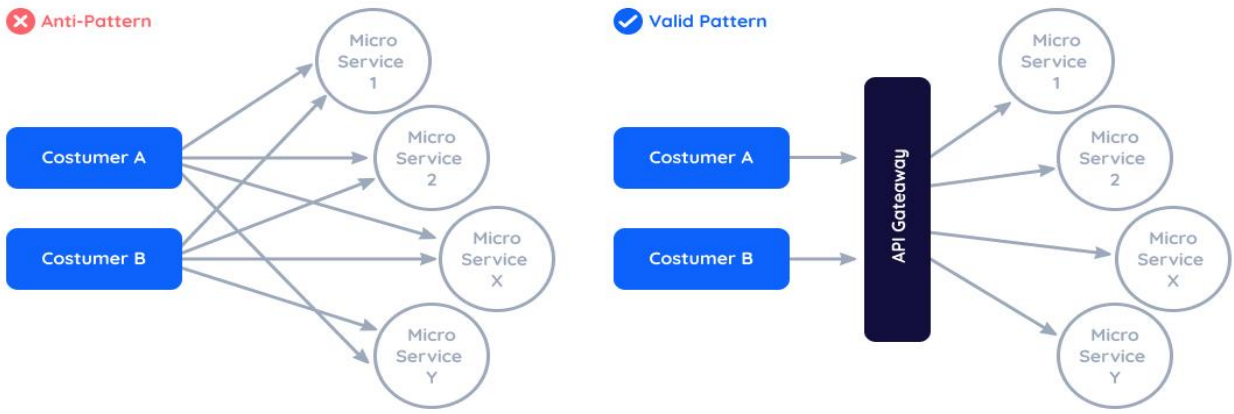
Experimental or behavioral:

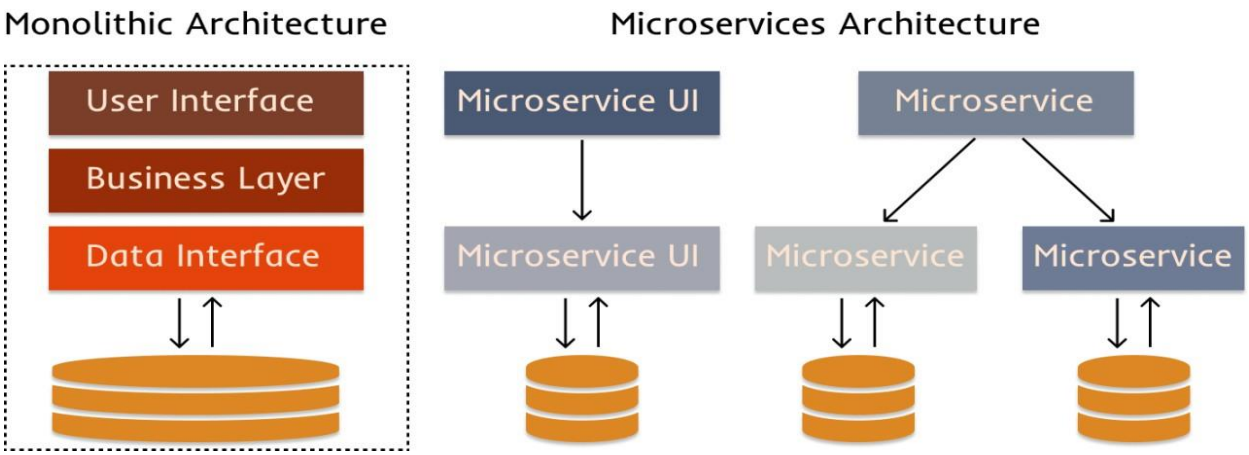
- **AIDetection JavaScript heuristic tool, keystroke dynamics, DroidDetect.** (arXiv)

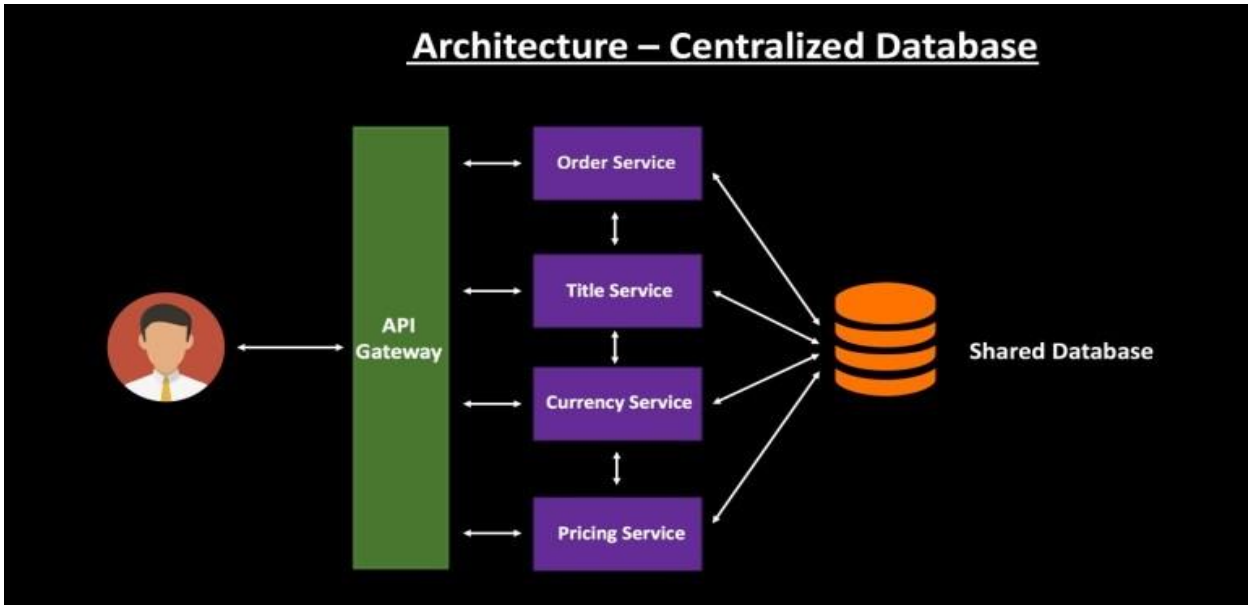
 Notes

- **Accuracy varies:** Claims of "99% accuracy" come from vendor testing — real performance differs by content type and domain. (Winston AI)
- **No single tool is definitive:** Combining tools (text + plagiarism + code + image) yields better coverage.
- **Metadata & watermark advances:** Tools like Google's SynthID use **embedded watermarks** to mark AI content at creation — powerful for future verification. (The Verge)
- **Ethics:** Many tools are used in **education and business policies**; false positives can have real consequences.

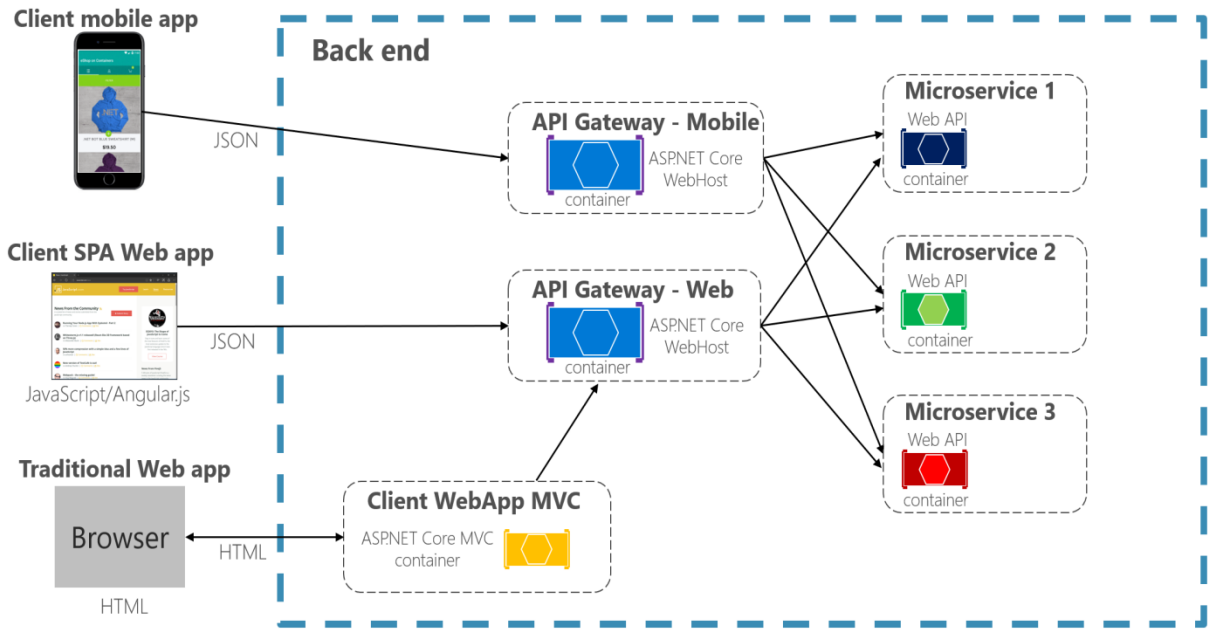
Below is an **expanded, architecture-grade Anti-Pattern → Correct-Pattern mapping table (35+ rows)**.
This is **print-friendly**, **interview-ready**, and **usable directly in design reviews**.







Using multiple **API Gateways / BFF**



✕ Anti-Pattern → ✓ Correct-Pattern Mapping (Master Table)

#	Anti-Pattern	What Goes Wrong	Correct Pattern	Why It Fixes the Problem
1	Distributed Monolith	Tight coupling across services	Decompose by Business Capability	True service autonomy
2	Shared Database	Schema & release coupling	Database per Service	Independent evolution
3	God Service	Single point of failure	Single Responsibility Service	Scales & changes safely
4	Chatty Services	High latency	API Gateway / Aggregator	Fewer network calls
5	Synchronous Everything	Cascading failures	Async Messaging	Loose coupling
6	Hardcoded Configuration	Redeploy for config change	Externalized Configuration	Runtime flexibility

7	Manual Service Discovery	Broken scaling	Service Registry	Dynamic discovery
8	Fixed IP / Hostnames	Cloud-incompatible	DNS / Service Discovery	Elastic infrastructure
9	Two-Phase Commit	Poor scalability	Saga Pattern	Distributed consistency
10	Central Shared Model	Domain leakage	Bounded Context (DDD)	Clean domain isolation
11	Direct DB Access Across Services	Data corruption	API-based Access	Ownership enforcement
12	No Failure Handling	Cascading outages	Circuit Breaker	Failure isolation
13	Infinite Retries	Resource exhaustion	Retry + Backoff + Timeout	Controlled recovery
14	No Timeouts	Thread blocking	Timeout Pattern	Resource protection
15	Single Thread Pool	Blast radius	Bulkhead Pattern	Fault isolation
16	No Fallbacks	Hard failures	Fallback Pattern	Graceful degradation
17	One API for All Clients	Over/under fetching	Backend for Frontend (BFF)	Client-optimized APIs
18	Direct Client-to-Service Calls	Security & coupling	API Gateway	Central control
19	Big-Bang Deployment	High production risk	Blue-Green Deployment	Zero downtime
20	Full Rollout at Once	Massive failures	Canary Deployment	Risk containment
21	Mutable Infrastructure	Configuration drift	Immutable Infrastructure	Predictable deploys
22	No Observability	Blind debugging	Centralized Logging	Faster root cause

23	Missing Request Tracking	Hard tracing	Distributed Tracing	End-to-end visibility
24	No Correlation IDs	Log chaos	Correlation ID Pattern	Traceability
25	No Health Checks	Slow recovery	Health Check API	Auto-healing
26	Custom Authentication per Service	Security bugs	OAuth2 / JWT	Standardized auth
27	Plain HTTP Between Services	Security risk	Mutual TLS (mTLS)	Strong trust
28	Secrets in Code	Credential leaks	Secrets Management	Secure storage
29	Direct Legacy Calls	Domain pollution	Anti-Corruption Layer	Clean boundaries
30	Full Legacy Rewrite	High failure risk	Strangler Fig Pattern	Gradual migration
31	Polling for Updates	Resource waste	Event-Driven Architecture	Efficient communication
32	Overloaded Read Queries	Performance issues	CQRS	Optimized reads
33	Overwriting Data State	Lost history	Event Sourcing	Full audit trail
34	Custom Networking Logic	Complex ops	Service Mesh	Built-in resilience
35	No Rate Limits	System overload	Rate Limiting	Traffic protection
36	Feature Changes via Deploy	Slow experiments	Feature Toggles	Runtime control
37	Monolithic CI/CD Pipeline	Slow releases	Independent Pipelines	Faster delivery

One-Line Interview Summary

“Every microservices anti-pattern is a symptom of coupling—correct patterns exist to restore autonomy, resilience, and scalability.”